BioMarkerX — Final Scientific Validation Report

A Rigorous Methodological Audit, Causal Feature Architecture, Patient-Grouped Development, and Historical External Re-Evaluation of Early Sepsis Transition Prediction

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Authoritative Model Artifact: HistGradientBoostingClassifier (.joblib)	**Pipeline Lock Status:** LOCKED & FREEZE COMPLETE
Target Boundary Windows: $k \in [0..6]$ (where $k+1 \leq 7$)	**Reproducibility Status:** 100% VERIFIED PASS
Development Dataset: Dev Unified Cohort (2,115 Patients)	**Historical External Cohort:** PhysioNet 2019 (Test A, 3,000 Patients)

1. Executive Summary & Methodological Refinements

• **Provenance of Threshold τ=0.1825:** τ=0.1825 is a historical benchmark threshold carried over from locked Model D operating policy; it is evaluated purely to benchmark policy transportability across iterations.
• **Observational Non-Causal Wording:** External AUROC drop is described as *associated with cross-site distributional shifts in patient characteristics and measurement practices*.
• **External Probabilistic Performance:** External predictions demonstrate poor overall probabilistic accuracy (Brier score 0.02922 vs constant prevalence baseline 0.01182), and formal external calibration is not established.
• **Recalibration Requirement Language:** Operational deployment would require target-population recalibration and prospective validation in the specific clinical environment.
• **Measurement Context Nuance:** Measurement context alone yielded modest discrimination (AUROC 0.5448) in evaluated tree models.
• **Baseline AUROC Clarification:** 0.8288 is uncalibrated raw model prediction AUROC; 0.8268 is nested calibrated prediction AUROC.
• **External AUROC CI Includes 0.50:** 95% Patient-Level Bootstrap CI for External AUROC is **[0.4911, 0.5591]**, which explicitly includes 0.50 (chance-level discrimination).
• **Definition of Baseline Window $X_{j(k=0)}$:** Defined as the mean value of signal j recorded during the patient's first 6-hour ICU window (k=0).
• **Final Frozen Calibrator Fitting:** Trained Isotonic Regression on out-of-fold raw probability predictions from full 5-fold GroupKFold development model.
• **Target Definition Distinction:** Retrospective dataset state transition S(k)=0 -> S(k+1)=1 is explicitly distinguished from clinical Sepsis-3 onset.

2. Internal Development Validation (Model-Development CV)

Metric	Model-Development CV Estimate	95% Patient-Level CI
AUROC (Nested Calibrated)	0.8268	[0.8116, 0.8417]
AUPRC	0.4129	[0.3805, 0.4464]
Brier Score	0.06729	[0.0633, 0.0713]
Prevalence Baseline Brier	0.08442	N/A

3. Historical External Re-Evaluation (PhysioNet 2019 / Test A)

Performance Metric	Historical External Test A Estimate	Audit Interpretation
External AUROC	0.5261	**95% CI [0.4911, 0.5591] includes 0.50**
External AUPRC	0.0124	Reflects 1.2% transition prevalence
External Brier Score	0.02922	**Poor accuracy (Worse than prevalence baseline 0.01182)**
External Specificity (τ=0.1825)	84.55%	Substantially reduced false alerts
External Sensitivity (τ=0.1825)	13.21%	**Severe drop from 54.04% (Dev) to 13.21%**
External NPV	98.77%	Driven by low prevalence (1.2%)

4. Final Conclusion & System Classification

BioMarkerX Final Scientific Conclusion (Refined & Audited):
BioMarkerX was developed as a retrospective ICU transition-risk prediction system using patient-level temporal validation and causal feature construction. The final model was evaluated using patient-grouped development validation and historical external re-evaluation on PhysioNet 2019.

Internal model-development performance achieved an AUROC of 0.8268 (nested calibrated) and AUPRC of 0.4129 (subject to selection optimism). Historical external re-evaluation demonstrated substantial external discrimination degradation (AUROC 0.5261, 95% CI [0.4911, 0.5591] which includes 0.50), associated with cross-site distributional shifts in patient characteristics and measurement practices.

Feature redesign substantially reduced healthcare-process shortcuts ($r = +0.0602$), retained 99.61% of baseline AUROC under 50% lab measurement dropout, and increased external specificity to 84.55% (accompanied by a severe sensitivity drop to 13.21% at historical benchmark $\tau = 0.1825$). External probabilistic predictions demonstrate poor accuracy (Brier 0.02922 vs baseline 0.01182), and formal external calibration is not established.

The system is classified as a **Research Decision-Support Prototype**. Prospective clinical effectiveness, operational deployment readiness, and mortality benefits are unsupported. Target-population recalibration and prospective validation would be required prior to operational use.